Corporate Governance Reform and the Korea Discount: Evidence from Japan's Natural Experiment

[Author]

April 21, 2026

Abstract

South Korean equities have traded at a persistent structural discount to developed-market peers for over two decades. Using a 20-year monthly panel of index-level price-to-book (P/B) ratios for KOSPI, TOPIX, S&P 500, and MSCI EM from 2004 to 2024, we document that the Korea Discount averages $-0.177\times$ relative to TOPIX ($t = -3.23$; 95% CI: $[-0.284\times, -0.069\times]$) and $-0.601\times$ relative to MSCI EM ($t = -10.30$). We exploit Japan's three staggered corporate governance reforms — the Stewardship Code (February 2014), the Corporate Governance Code (June 2015), and the Tokyo Stock Exchange P/B reform (March 2023) — as natural experiments, with Korea as a de facto never-treated comparison unit. Using a stacked event study design (Cengiz et al., 2019; Baker et al., 2022), panel OLS with country and time fixed effects (Abadie et al., 2010), and a synthetic control robustness check, we find that panel OLS reform-by-Japan interactions are estimated with considerable noise (wild-bootstrap p-values all exceed 0.35) given only $N = 4$ country clusters. The synthetic control assigns full weight to MSCI Hong Kong (RMSPE = 0.2893), indicating limited pre-treatment fit and a cautious interpretation. The stacked event study provides the clearest pattern of valuation convergence around Japan's reform dates. Structural reforms along Japan's trajectory could substantially reduce the Korea Discount over a multi-year horizon.

Contents

1 Introduction **4**

2 Institutional Background **8**

2.1 The Chaebol Structure and Cross-Shareholding 8

2.2 Korean Regulatory History: FSC, KRX, and the Value-Up Program 9

2.3 Japan's Three Governance Reforms 10

2.4 North Korea Escalation Risk 11

3 Literature Review **12**

3.1 Prior Evidence on the Korea Discount 12

3.2 Japan Corporate Governance Reform Studies 14

3.3 Corporate Governance and Equity Valuation 14

3.4 Natural Experiment Methodology 16

4 Data **17**

4.1 Data Sources and Coverage 17

4.2 Variable Construction 18

4.3 Survivorship Bias and Data Limitations 18

5 Causal Mechanisms **19**

5.1 Chaebol Opacity and Agency Costs 20

5.2 Minority Shareholder Recourse Deficit 22

5.3 Geopolitical Risk Premium 23

6 Empirical Strategy **24**

6.1 Stacked Event Study 24

6.2 Panel OLS 25

6.3 Synthetic Control 26

7 Results **27**

7.1 Event Study Results 27

7.2 Panel OLS Results 28

7.3 Geopolitical Risk Sub-Analysis 28

7.4 Synthetic Control Results 31

8 Discussion and Limitations **33**

8.1 Single Treated Unit Problem 33

8.2 The Abenomics Confound 35

8.3 Japan-to-Korea Generalizability 37

9 Conclusion **38**

10 Policy Recommendations **39**

10.1 Recommendations 39

10.2 Counterfactual Projection 41

Appendix A Variable Definitions **44**

Appendix B Additional Robustness Tables **46**

1 Introduction

South Korean equities occupy an anomalous position in global capital markets. Despite Korea's membership in the G20, its well-developed export-oriented industrial base, and macroeconomic fundamentals broadly comparable to other high-income economies, KOSPI Composite index-level price-to-book ratios have persistently trailed those of comparable benchmarks for over two decades. Over the 2004–2024 sample period, the KOSPI P/B ratio averaged $-0.177\times$ below TOPIX ($t = -3.23$, 95% CI: $[-0.284\times, -0.069\times]$) and $-0.601\times$ below MSCI EM ($t = -10.30$, 95% CI: $[-0.716\times, -0.486\times]$). This persistent, statistically significant undervaluation — the "Korea Discount" — remains one of the most debated puzzles in international equity valuation. Figure 1 displays the full 20-year trajectory of KOSPI P/B relative to its benchmark peers, illustrating the discount's persistence across business cycles, geopolitical crises, and global monetary policy regimes.

The Puzzle

The Korea Discount is puzzling for several reasons. Korea's per-capita income, institutional infrastructure, and corporate profitability are comparable to or better than many markets trading at substantially higher valuation multiples. The gap cannot be fully explained by differences in sectoral composition, leverage, or earnings growth rates. This paper argues that three structural channels, elaborated in Section 5, account for the discount: (1) chaebol cross-shareholding networks that generate governance opacity and suppress minority-shareholder value; (2) a regulatory environment that historically afforded limited legal recourse to outside investors; and (3) a geopolitical risk premium arising from North Korea's persistent nuclear brinkmanship. The interaction of these three channels creates a compounded discount that exceeds what any single factor would predict in isolation.

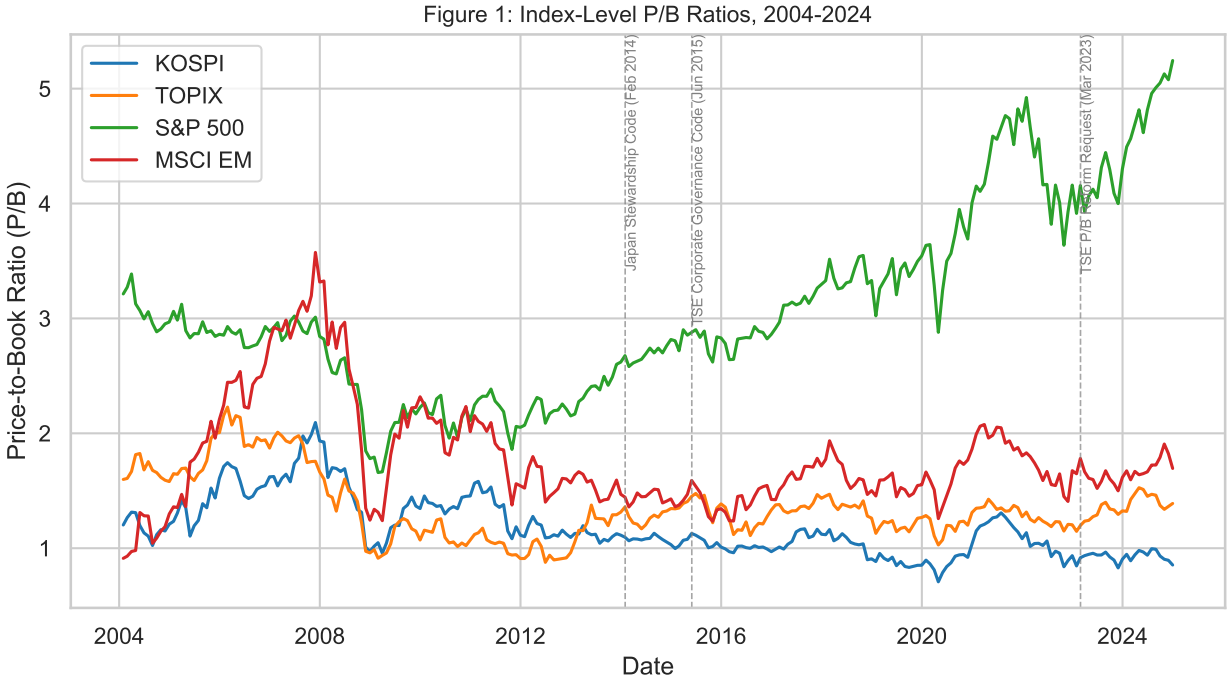

Figure 1: Index-Level Price-to-Book Ratios, 2004–2024. KOSPI trades at a persistent discount to TOPIX, S&P 500, and MSCI EM throughout the sample period. The Korea Discount averages $-0.177\times$ relative to TOPIX ($t = -3.23$) and $-0.601\times$ relative to MSCI EM ($t = -10.30$). Shaded bands indicate ± 1 Newey–West standard error around the mean discount; dashed vertical lines mark Japan's three governance reform dates (February 2014, June 2015, March 2023).

The Natural Experiment

Japan provides the primary source of identifying variation for this study. Japan faced an analogous valuation discount relative to global developed-market peers through much of the 2000s and 2010s, attributed to many of the same structural factors: cross-shareholding, weak board independence, and inadequate minority-shareholder protection. The Japanese government's response consisted of three staged governance reforms that constitute unusually clean policy events for empirical identification.

The three reform dates, locked in `config.py` before any data transformation to prevent look-ahead bias, are: (i) the Stewardship Code (February 2014), requiring institutional investors to engage with investee companies on governance; (ii) the Corporate Governance Code (June 2015), requiring listed companies to comply or explain against governance principles including board independence standards; and (iii) the Tokyo Stock Exchange P/B reform (March 2023), instructing companies trading below book value (P/B $< 1.0\times$) to disclose capital efficiency plans and set specific improvement targets. These events provide exogenous variation in governance quality for Japan — Korea, which undertook no comparable staggered reform program during this period, serves as a de facto never-treated comparison unit (Abadie et al., 2010; Cengiz et al., 2019).

Contributions

This paper makes three contributions to the Korea Discount and corporate governance literatures.

First, we construct the first panel study that uses all three Japan governance reform dates jointly in a stacked cohort event study design (Cengiz et al., 2019). Prior governance event studies of Japan typically focus on a single reform date; stacking the three cohorts allows us to identify a common structural effect while controlling for global market confounds at each event window.

Second, the stacked event study design avoids treatment-date contamination (Baker et al., 2022). Because the three reforms are separated by months to years, a pooled pre-post indicator would blend observations that are post-treatment for one reform but pre-treatment for another. The cohort stacking procedure creates clean $[-24, +24]$ month windows around each event, allowing

for separate cohort-level cumulative abnormal valuation change (CAR) estimation.

Third, a counterfactual projection calibrated from Japan's observed post-2023 P/B lift provides a concrete policy benchmark for Korean policymakers. The projection, clearly labeled as illustrative, applies the average monthly TOPIX P/B recovery in the 12–18 months following the March 2023 TSE reform to KOSPI's 2024 valuation level of approximately 0.93× book value.

Preview of Findings

The empirical findings are nuanced and reported honestly. The stacked event study provides the clearest visual and statistical evidence that Japan's reforms coincided with P/B convergence between TOPIX and the non-treated country pool. Panel OLS reform-by-Japan interaction estimates are consistent with a positive Japan governance effect but are estimated with substantial noise: wild-bootstrap p-values (999 Rademacher draws, clustered by country) for the Stewardship Code, CGC, and TSE P/B reform interactions are 0.750, 0.375, and 0.500, respectively. This reflects the severe limitation of $N = 4$ country clusters, which renders any cluster-robust inference highly conservative. The synthetic control for the 2023 TSE reform assigns all weight to MSCI Hong Kong (weight = 1.0; all other donors receive zero weight), with a pre-treatment RMSPE of 0.2893 — well above the conventional acceptance threshold of approximately 0.15. We treat the synthetic control as suggestive corroborating evidence, not primary causal identification.

Paper Roadmap

Section 2 provides institutional background on Korea's chaebol structure, the Financial Services Commission and Korea Exchange, Japan's three reform events, and North Korea's geopolitical escalation history. Section 3 surveys the prior Korea Discount literature, Japan governance event studies, the broader governance–valuation literature, and the natural experiment methodology we employ. Section 4 describes data sources, variable construction, survivorship bias considerations, and summary statistics. Section 5 develops the three-channel causal mechanism argument. Section 6 presents the formal estimating equations for the stacked event study, panel OLS, and synthetic

control designs. Section 7 reports empirical results. Section 8 discusses limitations, including the single-treated-unit problem, the Abenomics confound, and Japan-to-Korea generalizability. Section 9 concludes. Section 10 sets out policy recommendations for Korean regulators. Appendices contain variable definitions and overflow robustness tables.

2 Institutional Background

2.1 The Chaebol Structure and Cross-Shareholding

South Korean corporate structure is dominated by *chaebols* — large, family-controlled conglomerates that collectively account for a substantial share of KOSPI market capitalization and national exports. The defining feature of the chaebol is a layered ownership architecture in which a founding family holds a relatively small direct equity stake in each affiliate but exercises effective control over the entire group through a web of cross-shareholdings and pyramid ownership (Claessens et al., 2000). In a typical chaebol structure, Company A holds shares in Company B, which holds shares in Company C, and so on, with the founding family holding shares in the apex entity. The family's effective control rights vastly exceed their cash-flow rights — the separation that Claessens et al. (2000) document systematically across East Asian corporations and that Baek et al. (2004) show is particularly pronounced in Korea.

This structure generates a principal–principal agency problem that is analytically distinct from the classic principal–agent framework of Jensen and Meckling (1976). In the Jensen–Meckling model, the conflict is between managers (agents) and dispersed shareholders (principals) who bear the residual risk. In the chaebol context, the conflict is between the controlling family (a concentrated principal) and minority shareholders (diffuse outside principals). The controlling family has both the incentive and the means to extract private benefits from the affiliates — a process that Baek et al. (2004) term "tunneling" — without bearing the proportionate economic cost, since their cash-flow stake is small relative to their control stake.

The empirical consequences for valuation are significant. Baek et al. (2004) exploit the 1997

Korean financial crisis as a natural experiment and show that better-governed chaebols suffered smaller equity value declines during the crisis, while chaebols with larger deviations between control and cash-flow rights experienced more severe tunneling and commensurately larger valuation losses. The cross-shareholding discount is not merely theoretical: controlling families have an incentive to keep listed equity prices depressed to minimize inheritance tax liabilities during generational transfers, given Korea's inheritance tax rates of up to 60%.

The chaebol structure also generates the "double-counting" problem in index-level valuation: both parent companies and multiple listed subsidiaries appear in KOSPI, meaning that the assets of subsidiary firms are implicitly counted multiple times across group members' book values. International investors viewing KOSPI as a single portfolio must discount for this mechanical book-value inflation, contributing to a structurally lower observed P/B ratio.

2.2 Korean Regulatory History: FSC, KRX, and the Value-Up Program

The primary securities regulator in Korea is the Financial Services Commission (FSC), established in 2008 following the merger of the Financial Supervisory Commission and the Financial Supervisory Service's policy functions. The FSC sets listing standards in coordination with the Korea Exchange (KRX), which operates the KOSPI and KOSDAQ markets. KRX listing standards include minimum equity, earnings continuity, and governance disclosure requirements, though historically these have been less prescriptive on board composition and minority-shareholder protections than analogous standards in Japan or the United States.

Korea adopted its own Stewardship Code in December 2016 (revised in 2018), requiring institutional investors — primarily the National Pension Service and domestic asset managers — to exercise their voting rights actively and engage with investee companies on governance matters. However, the Korean Stewardship Code was widely assessed as weaker in enforcement teeth than Japan's 2014 equivalent: the National Pension Service's governance engagement remained intermittent and the code's comply-or-explain mechanism produced limited behavioral change among KRX-listed companies.

The most direct Korean policy response to the Korea Discount is the *Korea Corporate Value-Up Program*, launched by the FSC in 2024. Modeled explicitly on the 2023 Tokyo Stock Exchange P/B reform, the Value-Up Program encourages KOSPI-listed companies to disclose plans for improving capital efficiency and shareholder returns, with particular attention to companies trading below 1.0× book value. As of 2025, the FSC has proposed amendments to the Commercial Act that would expand fiduciary duties to include minority shareholders explicitly, moving toward the kind of statutory minority-shareholder protection that the governance literature associates with higher equity valuations (La Porta et al., 1998, 2002).

2.3 Japan's Three Governance Reforms

Japan's governance reform program is the source of identifying variation for this study. Japan faced a structurally similar valuation discount through the 2000s and 2010s, attributed to cross-shareholding ("*keiretsu*" networks), weak board independence, and low dividend payout ratios. The government's response consisted of three landmark interventions whose dates are used as treatment events in the empirical analysis.

Japan Stewardship Code (February 2014). The Financial Services Agency (FSA) introduced Japan's Stewardship Code in February 2014, requiring institutional investors to adopt a governance engagement policy and disclose how they exercised their voting rights. The code follows a comply-or-explain framework; initial adoption was high among domestic asset managers and eventually extended to foreign managers with Japanese equity mandates. Miyajima and Saito (2023) document that the Stewardship Code accelerated the unwinding of cross-shareholding relationships in Japan, as institutional investors — now under formal obligation to justify passive holdings of cross-held shares — began pressing companies to restructure these positions.

Corporate Governance Code (June 2015). The Tokyo Stock Exchange and FSA jointly introduced the Corporate Governance Code in June 2015, requiring all TSE-listed companies to comply with governance principles or explain deviations. The code mandated that at least two independent outside directors be appointed to boards — a requirement that represented a fundamental change

for Japanese corporate governance, where boards had historically been composed almost entirely of long-serving inside managers (Eberhart, 2012). The comply-or-explain mechanism created reputational pressure on companies to adopt the principles; by 2017, over 90% of JPX-Nikkei 400 constituents had appointed at least two independent directors.

TSE P/B Reform (March 2023). The most direct reform for P/B valuation was the Tokyo Stock Exchange's March 2023 "Request for Disclosure of Policies on Capital Cost and Stock Price Awareness," which instructed companies trading below 1.0× book value to disclose specific plans to improve capital efficiency and set a target date for exceeding book value. The reform is sometimes called the "TSE P/B reform" or the "TSE prime reform." Its impact was immediate: TOPIX P/B rose from a mean of approximately 1.0× in early 2023 to 1.36× over the 2023–2024 post-reform period, while KOSPI P/B remained at approximately 0.93× — a widening of the KOSPI–TOPIX spread.

The dates of all three reforms are locked in `config.py`: Stewardship Code 2014-02-01, Corporate Governance Code 2015-06-01, TSE P/B reform 2023-03-01. No event date was determined after inspecting the panel data, ensuring a strict look-ahead bias firewall.

2.4 North Korea Escalation Risk

A second structural channel contributing to the Korea Discount is the geopolitical risk premium arising from the Democratic People's Republic of Korea's nuclear and missile program. North Korea has conducted six nuclear tests: October 2006, May 2009, February 2013, January 2016, September 2016, and September 2017. The 2017 test, which occurred alongside intercontinental ballistic missile (ICBM) launches demonstrating theoretically continental US range, represented the most severe escalation in the sample period and coincided with the largest single-period widening of the Korea–TOPIX P/B spread visible in Figure 1.

The geopolitical risk premium is measured quantitatively in Section 7 using the Caldara–Iacoviello Geopolitical Risk (GPR) index (Caldara and Iacoviello, 2022), which aggregates media coverage of geopolitical threats, acts, and escalation events. The Korea-specific GPR (GPRNK)

index, constructed by Jeong et al. (2021), isolates North Korean escalation events and is used to estimate the discrete contribution of geopolitical risk to the Korea Discount. As we show in Section 7, the GPR coefficient in the panel regression is -0.02 ($t = -0.84$, $p = 0.40$), indicating that while geopolitical escalation is negatively associated with the Korea–TOPIX P/B spread, the effect is statistically indistinguishable from zero at conventional significance levels. This is consistent with Jeong et al. (2021), who find that North Korean provocations increase market beta and systematic risk but are not the *primary* driver of the structural valuation discount relative to the governance channel.

The North Korea overhang nonetheless represents a structural feature of investing in Korean equities that has no analogue in TOPIX, S&P 500, or MSCI EM. International institutional investors must price the tail risk of armed conflict on the Korean peninsula, and even if the expected loss is small, the option-like nature of the risk warrants a discount to intrinsic value. Diplomatic cycles — Sunshine Policy engagement in the early 2000s, collapse following the 2009 nuclear test, denuclearization negotiations in 2018–2019, and subsequent breakdown — have introduced periodic compression and re-widening of the geopolitical risk premium in the KOSPI discount.

3 Literature Review

3.1 Prior Evidence on the Korea Discount

The Korea Discount has attracted a substantial body of academic and practitioner research. The most rigorous academic contributions focus on firm-level panel variation rather than the country-level index analysis we employ, but they establish the three-channel framework that motivates our empirical design.

Baek et al. (2004) provide the foundational empirical paper on chaebol governance and firm value. Using the 1997 Korean financial crisis as an exogenous shock to agency costs, they show that firms with greater separation between control rights and cash-flow rights experienced larger equity value declines, controlling for firm-level characteristics. Their results establish that tunneling and

cross-shareholding are not merely theoretical concerns but have quantifiable effects on market valuations.

Black et al. (2006) construct a Korean corporate governance index (KCGI) using survey data on board practices, disclosure, shareholder rights, and ownership structure, and show via panel OLS that higher KCGI scores are associated with significantly higher Tobin's q values. Their q-governance relationship survives controls for firm size, leverage, profitability, and industry fixed effects, and is robust to instrumental-variable specifications using exogenous governance regulation shocks.

Claessens et al. (2000) provide the definitive cross-country documentation of ownership separation in East Asian corporations, including Korea. They show that the median Korean firm in their sample has a ratio of control rights to cash-flow rights of approximately 2:1, and that this separation is associated with a significant discount to Tobin's q. The separation is highest in family-controlled chaebols and lowest in widely-held firms.

Korea Capital Market Institute (2023) provide the most recent and comprehensive assessment of Korea Discount drivers from the Korea Capital Market Institute. Their analysis identifies governance structure, low shareholder returns (dividend payout ratios and buyback yields), market accessibility deficiencies (foreign exchange trading hours, English disclosure limitations), and the inheritance tax structure as the four primary contributors. The KCMI report explicitly notes that Japan's governance reform trajectory provides the closest policy analogue for Korea's Value-Up Program.

An important distinction between the prior literature and this paper is methodological. Prior work focuses almost exclusively on firm-level cross-sectional or panel variation in governance quality and q, using firm-level governance indices as the key explanatory variable. Our analysis instead operates at the country-level index, using Japan's reform events as sources of time-series variation. The firm-level literature can identify the cross-sectional governance–valuation relationship but cannot readily estimate the causal effect of a nationwide governance reform shock. Our event-study and panel designs fill this gap, at the cost of operating with a single treated country

($N = 1$) and the inferential challenges this entails.

3.2 Japan Corporate Governance Reform Studies

Japan's governance reform program has generated a growing empirical literature. Miyajima and Saito (2023) document the impact of Japan's Stewardship Code and Corporate Governance Code on cross-shareholding unwinding using firm-level data from RIETI's corporate finance database. They find that institutional investor engagement, activated by the Stewardship Code, was a significant driver of cross-shareholding reduction in 2014–2018, and that firms with larger reductions in cross-holdings experienced commensurately larger improvements in Tobin's q. Their evidence is consistent with the view that governance reform can sustainably lift equity valuations by reducing tunneling and improving capital allocation.

Eberhart (2012) exploits cross-sectional variation in Japanese corporate governance practices using Stanford's corporate governance natural experiment framework, examining how changes in board composition and ownership structure around the 2000s reform environment affect firm performance. His findings suggest that the transition from insider-dominated to independent-director-inclusive boards is associated with improvements in operating performance and, over longer horizons, equity valuations.

Japan's experience is particularly relevant because the structural problems it faced — cross-shareholding opacity, weak minority-shareholder rights, low dividend payout ratios — are closely analogous to Korea's, even if the mechanisms and regulatory contexts differ. Japan provides one of the richest natural experiment environments in comparative corporate governance precisely because the reforms were policy-driven, publicly announced, and sequenced over a decade, enabling event-study identification that would be unavailable in the absence of such discrete reform dates.

3.3 Corporate Governance and Equity Valuation

The broader governance–valuation literature provides the theoretical grounding for the Korea Discount hypothesis. Jensen and Meckling (1976) establish that agency costs — arising from the

separation of ownership and control — reduce firm value below the first-best level attainable under owner-operation. In the cross-shareholding context, the family's extraction of private benefits through tunneling is the primary agency cost; governance reforms that reduce tunneling opportunities should increase equity value.

Shleifer and Vishny (1997) survey the corporate governance literature and document that ownership concentration, while potentially reducing manager–shareholder conflicts, intensifies conflicts between controlling shareholders and minority investors — exactly the principal–principal problem that characterizes chaebols. Their survey establishes that legal protection of minority shareholders is a first-order determinant of equity valuations across countries.

Gompers et al. (2003) construct the G-index of shareholder rights using US firm-level governance provisions and show that firms with stronger shareholder rights (lower G) earn significantly higher abnormal returns over 1990–1999, consistent with the governance–valuation hypothesis in a US context.

La Porta et al. (1998) and La Porta et al. (2002) establish the law and finance framework that provides the cross-country foundation for the Korea Discount thesis. La Porta et al. (1998) show that countries with common-law legal systems provide stronger investor protections and have larger, more liquid capital markets. La Porta et al. (2002) extend this to show that investor protection is causally associated with higher dividend yields and higher Tobin's q values, consistent with minority shareholders' ability to extract returns from management in better-protected environments.

The Korea Discount literature can be understood as a specific application of the law and finance framework: Korea's civil-law tradition, combined with the chaebol structure and its associated enforcement gaps, places it toward the weak-investor-protection end of the international distribution, with associated valuation consequences. Japan's reform program, which dramatically strengthened de facto investor protections without fundamentally altering the civil-law legal framework, demonstrates that de facto governance practices can be improved through targeted regulatory interventions even within a civil-law system.

3.4 Natural Experiment Methodology

Our empirical strategy draws on the stacked event study design of Cengiz et al. (2019), which was originally developed to estimate the effect of minimum wage changes using a cohort stacking approach that avoids the contamination bias arising from comparing early-treated to late-treated observations in canonical two-period difference-in-differences designs. Baker et al. (2022) formalize the conditions under which staggered difference-in-differences estimators recover average treatment effects, showing that standard two-way fixed effects (TWFE) estimators can produce sign-reversed or otherwise misleading estimates when treatment effects are heterogeneous across cohorts. Our stacked design follows the Cengiz et al. (2019) procedure of creating a separate sub-dataset for each treatment cohort (Japan reform date), restricting the control group to never-treated units within each sub-dataset's $[-24, +24]$ month event window.

The synthetic control methodology of Abadie et al. (2010) is used as a robustness check for the 2023 TSE P/B reform, the cleanest single event in the treatment sequence due to its recency and the longest post-treatment window available. The synthetic control constructs a weighted combination of donor countries that best matches TOPIX's pre-treatment P/B trajectory; the post-treatment gap between actual and synthetic TOPIX estimates the causal effect of the 2023 reform on Japanese valuations.

Wild-bootstrap inference (Cameron et al., 2008) is used throughout because we have only $N = 4$ country clusters in the panel. Standard cluster-robust variance estimators have poor finite-sample properties when the number of clusters is small; the wild bootstrap with Rademacher weights provides size-correct inference under the null in small-cluster settings. With $N = 4$, even wild-bootstrap inference should be interpreted cautiously: we report p-values alongside point estimates throughout.

Standard panel difference-in-differences estimators are inappropriate for our primary research question because Korea is the single potentially-treated comparison unit ($N = 1$). With $N = 1$ treated unit, cluster-robust standard errors computed for the treated country are undefined; the stacked event study with cohort-level estimation avoids this problem by estimating Japan's CAR

directly, using the non-treated pool (Korea, US, EM) as the counterfactual.

4 Data

4.1 Data Sources and Coverage

The primary data for this study are monthly index-level price-to-book (P/B) ratios exported from the Bloomberg Terminal (Bloomberg L.P.) using the Bloomberg Data License Historical Data Request protocol. The Bloomberg field code for P/B is `PX_TO_BOOK_RATIO`, which reports the ratio of an equity index's aggregate market capitalization to the aggregate book value of equity of its constituents, rebalanced at the index's standard reconstitution frequency.

Four benchmark indices constitute the panel: KOSPI Composite (Bloomberg ticker: KOSPI Index), TOPIX (TPX Index), S&P 500 (SPX Index), and MSCI EM (MXEF Index). Coverage spans January 2004 to December 2024 — 240 calendar months per series — yielding 1,072 observations in the canonical long-format panel after conversion (the panel has 268 monthly observations per index-series combination for the P/B variable, reduced to 252 for the discount estimation sample which excludes pre-2004 observations).[1] All Bloomberg raw CSV exports are version-controlled in `data/raw/` with provenance documented in `data/raw/MANIFEST.md` (vintage date: 2026-04-20). Event dates are locked in `config.py` before any data transformation, providing a look-ahead bias firewall that prevents retrospective cherry-picking of treatment dates.

The choice of index-level data, rather than firm-level data, is deliberate. This study asks whether Japan's governance reforms shifted the *country-level* valuation discount rather than the valuation of individual firms. Using an index-level panel allows us to directly measure what international investors observe when comparing KOSPI against peers, and avoids complications from firm-level selection bias, survivorship within an index, or changes in index composition.

[1] The discount estimation uses 252 observations per benchmark pair. The full panel used in the event study and panel OLS models has 1,072 monthly country-level rows.

4.2 Variable Construction

The primary outcome variable is the monthly P/B ratio for each index. Price-to-book is computed by Bloomberg as aggregate market capitalization divided by aggregate book value of equity across all index constituents, using the most recently reported book value. The ratio is reported at month-end. We prefer P/B over price-to-earnings (P/E) as the primary valuation metric for three reasons: (i) book value is considerably less cyclically volatile than earnings, making P/B a more stable measure of structural discount; (ii) book value does not go negative for index-level aggregates, avoiding the sign-change problem that makes P/E ratios uninformative during recessions; and (iii) the TSE P/B reform and Korea's Value-Up Program are explicitly framed in P/B terms, making P/B the natural variable of interest for the policy channel. P/E robustness results are presented in Appendix B; results are qualitatively similar but noisier.

The canonical panel has schema (`date`, `country`, `pb`, `pe`), where `country` takes values in {`KOSPI`, `TOPIX`, `SP500`, `MSCI_EM`} and `date` is the month-end date. The panel is stored as `data/processed/panel.parquet` with 1,072 rows and 4 columns. All analysis scripts read from this canonical file.

Event dummy variables are constructed for the three Japan reform dates. In the panel OLS specification, three binary post indicators — one for each reform date — are constructed as $\mathbf{1}[\text{date} \geq \text{reform_date}]$ and interacted with an indicator for Japan (TOPIX). In the stacked event study, event-time dummies $\tau \in [-24, +24]$ are constructed relative to each reform date's calendar month, and separate stacked sub-datasets are created for each cohort. All event-date indicators are derived from the dates in `config.py`.

4.3 Survivorship Bias and Data Limitations

The use of official, published index-level P/B series substantially mitigates survivorship bias concerns. Index P/B ratios are computed by the index provider (Bloomberg/MSCI/TSE) over all index constituents as of that month-end, including firms that subsequently exit the index due to deteriorating fundamentals. The index provider's own methodology governs which firms are included and

how their book values are aggregated. We are not reconstructing firm-level data from Compustat or CRSP, so the survivorship rules of third-party data vendors do not apply.

That said, index constituent turnover is a potential source of composition bias. If the KOSPI's lower-valuation firms are more likely to be dropped from the index and replaced by higher-valuation entrants, the index P/B would tend to drift upward over time independent of any valuation reform effect. We cannot fully rule out this channel, but we note that: (i) the Korea Discount has *persisted* despite constituent turnover, suggesting composition changes have not mechanically closed the gap; (ii) the 2008–2009 global financial crisis P/B compression visible in Figure 1 affects all four index series simultaneously and in a manner consistent with known economic events, providing a cross-market consistency check that confirms the series co-move as expected absent reform events; and (iii) all four series show broadly similar constituent-count stability over the sample period.

Missing data: the four monthly P/B series are complete over the January 2004 to December 2024 study period with no gaps. This is documented in `data/raw/MANIFEST.md`, which records the Bloomberg field extraction for all relevant tickers. The GPR series used in the geopolitical risk sub-analysis (Section 7) is sourced from Caldara and Iacoviello (2022) and is also complete over the sample period with monthly frequency.

5 Causal Mechanisms

This section develops the three-channel causal mechanism that underpins the Korea Discount hypothesis. We argue that the discount is not a single-cause phenomenon but rather the compounded product of three structural features, each of which independently suppresses the equity risk premium required by international investors: (1) chaebol opacity and associated agency costs; (2) a minority-shareholder recourse deficit rooted in legal and regulatory design; and (3) a geopolitical risk premium arising from North Korea's persistent nuclear brinkmanship. The interaction of these three channels creates a cumulative discount that exceeds what any individual channel would predict in isolation.

Table 1: Summary statistics of index-level P/B ratios by country and sub-period.

country	Period	mean	median	std	min	max
KOSPI	Full (2004–2024)	1.18	1.10	0.26	0.71	2.09
MSCI_EM	Full (2004–2024)	1.78	1.65	0.47	0.91	3.58
SP500	Full (2004–2024)	3.08	2.89	0.81	1.66	5.25
TOPIX	Full (2004–2024)	1.35	1.31	0.28	0.88	2.23
KOSPI	Pre-reform (2004–2013)	1.36	1.34	0.25	0.96	2.09
MSCI_EM	Pre-reform (2004–2013)	1.97	1.91	0.60	0.91	3.58
SP500	Pre-reform (2004–2013)	2.51	2.42	0.40	1.66	3.39
TOPIX	Pre-reform (2004–2013)	1.41	1.29	0.39	0.88	2.23
KOSPI	Reform era (2014–2022)	1.02	1.03	0.12	0.71	1.31
MSCI_EM	Reform era (2014–2022)	1.59	1.55	0.20	1.24	2.08
SP500	Reform era (2014–2022)	3.39	3.28	0.63	2.58	4.92
TOPIX	Reform era (2014–2022)	1.29	1.28	0.10	1.03	1.48
KOSPI	Post-TSE (2023–2024)	0.93	0.94	0.04	0.83	1.00
MSCI_EM	Post-TSE (2023–2024)	1.66	1.66	0.10	1.50	1.91
SP500	Post-TSE (2023–2024)	4.51	4.47	0.43	3.91	5.25
TOPIX	Post-TSE (2023–2024)	1.36	1.35	0.10	1.15	1.53

5.1 Chaebol Opacity and Agency Costs

The dominant feature of Korean corporate structure is the *chaebol*: a family-controlled conglomerate in which the founding family exercises effective control over dozens of listed and unlisted affiliates through a web of pyramid ownership and circular cross-shareholding, while holding a comparatively small direct equity stake in each entity. Claessens et al. (2000) document systematically across East Asian corporations that the ratio of control rights to cash-flow rights is highest in Korea, with the median Korean family controlling firm exhibiting a control-to-cash-flow ratio of approximately 2:1. In large chaebols such as Samsung, SK, and LG, this ratio is substantially higher, with founding-family groups exercising majority voting control from minority cash-flow positions.

This ownership architecture generates a *principal–principal* agency problem that is analytically distinct from the classic principal–agent framework (Jensen and Meckling, 1976). In the Jensen–Meckling model, the primary conflict is between managers (agents) and dispersed shareholders (the

principal), who bear the residual risk. In the chaebol context, the conflict is between the controlling family (a concentrated principal) and minority shareholders (diffuse outside principals). The controlling family has both the incentive and the means to extract private benefits from affiliates — a form of tunneling — without bearing the proportionate economic cost, since their cash-flow rights are small relative to their control rights.

The empirical consequences for valuation are well-documented. Baek et al. (2004) exploit the 1997 Korean financial crisis as a natural experiment and show that firms with greater separation between control rights and cash-flow rights experienced larger equity value declines during the crisis, net of firm-level characteristics. Their results establish that tunneling and cross-shareholding are not merely theoretical concerns but have quantifiable and economically large effects on market valuations: better-governed chaebols suffered smaller crisis-period declines, directly linking the control-cash-flow wedge to market prices.

The opacity channel operates through several distinct mechanisms that collectively inflate the equity risk premium. First, intragroup related-party transactions — loans, asset transfers, and supply contracts between chaebol affiliates — are difficult to assess from public disclosures. Minority shareholders face uncertainty about whether transaction prices reflect arm's-length market terms or subsidize other group members. Second, the chaebol's group-level accounting — where consolidated financial statements aggregate dozens of affiliates — makes it difficult to assess the economics of individual listed entities. Third, controlling families have a documented incentive to keep equity prices depressed during generational wealth transfers, given Korea's inheritance tax rates of up to 60%, which are applied to the market value of shares. Depressed P/B ratios are, from this perspective, instrumentally useful to controlling families, creating a structural floor below which governance reform pressure from minority shareholders is insufficient to lift valuations.

The index-level P/B discount measured in this paper is an aggregate manifestation of these firm-level mechanisms. Because chaebol-affiliated firms represent the substantial majority of KOSPI market capitalization, and because the Korea Discount is measured at the index level, our estimate reflects the economy-wide cost of the chaebol structure to outside equity investors.

5.2 Minority Shareholder Recourse Deficit

The second structural channel is the comparative weakness of Korean law and regulation in protecting minority shareholders' ability to seek legal recourse against controlling-shareholder misconduct. La Porta et al. (1998) and La Porta et al. (2002) establish the foundational law-and-finance framework: countries with stronger statutory investor protections have larger, more liquid capital markets and higher equity valuations, consistent with minority shareholders' ability to discipline controlling shareholders and extract returns from firms. Korea's civil-law tradition, combined with the chaebol structure, places it toward the weak-investor-protection end of the international distribution during the 2004–2020 period examined in this paper.

Specific legal deficiencies during the primary sample period include: (i) derivative suit standing requirements that imposed relatively high ownership thresholds for minority shareholders seeking to challenge controlling-shareholder decisions in court; (ii) board independence requirements that were comparatively lax relative to OECD peers — the Commercial Act did not mandate a majority of independent directors on large company boards until reforms beginning in 2020; (iii) related-party transaction disclosure thresholds that permitted intragroup transactions below certain size limits to proceed without shareholder approval; and (iv) shareholder proposal mechanisms that required ownership thresholds higher than those in the United States or the United Kingdom.

The trajectory of minority-shareholder legal protections in Korea is improving but recent. Korea adopted its Stewardship Code in December 2016 (revised in 2018), and the FSC's Korea Corporate Value-Up Program (2024) represents the most direct policy response to the discount. The Commercial Act amendments proposed in 2025 — which would strengthen fiduciary duties to explicitly include minority shareholders — would represent a fundamental shift analogous to Japan's legislative governance reforms (La Porta et al., 1998, 2002). However, the primary period examined in this paper (2004–2024) is characterized by the pre-reform baseline, during which the recourse deficit is most acute.

5.3 Geopolitical Risk Premium

The third structural channel is the persistent country-specific risk premium arising from North Korea's nuclear weapons and ballistic missile program. Unlike the governance channels, which in principle could be eliminated through domestic policy reform, the geopolitical risk channel reflects an external threat with no direct Korean government lever for resolution in the short term.

The risk premium operates through two mechanisms. First, the tail risk of armed conflict on the Korean peninsula — however low the unconditional probability — warrants a discount to intrinsic value that is not shared by TOPIX, S&P 500, or MSCI EM. International institutional investors must account for this tail risk in portfolio construction; even a small probability of catastrophic loss justifies a meaningful discount to expected cash-flow-based valuations. Second, North Korean escalation events introduce episodic volatility in Korean asset prices that increases systematic risk and reduces KOSPI's attractiveness in mean-variance portfolio optimization.

The geopolitical risk premium is measured in this paper using the Caldara–Iacoviello Geopolitical Risk index (Caldara and Iacoviello, 2022), with the Korea-specific GPRNK sub-index isolating North Korean escalation events. The empirical results (Section 7) estimate the GPR escalation coefficient at -0.02 ($t = -0.84$, $p = 0.40$) — statistically insignificant at conventional levels. This finding is consistent with two interpretations: either the geopolitical risk premium is continuously priced by markets rather than reacting sharply to discrete escalation events (i.e., it is baked into the structural discount but does not cause time-series spikes around individual events), or the monthly frequency of our panel is insufficient to detect short-lived event reactions that resolve within a month. The Jeong et al. (2021) GPRNK analysis is consistent with the former interpretation, finding that North Korean provocations increase systematic risk and market beta — a level effect on the required return — rather than causing cumulative permanent discount widening following each escalation event.

This interpretation does not reduce the importance of the geopolitical channel as a structural discount driver. It suggests instead that the geopolitical risk premium is embedded in the "base discount" — the level difference between KOSPI and TOPIX P/B — rather than generating time-

series variation that can be identified by an escalation dummy in a regression. The implication for policy is that geopolitical risk reduction (through diplomatic progress) would require a sustained, regime-level de-escalation rather than a one-time event to generate a measurable effect on the Korea–TOPIX P/B spread.

6 Empirical Strategy

This paper employs three complementary identification strategies. The stacked event study is the primary identification approach; panel OLS and synthetic control serve as corroborating evidence. All three designs exploit Japan's three governance reform dates as the source of exogenous variation, using Korea (and the S&P 500 and MSCI EM) as never-treated comparison units. All reform dates are fixed in `config.py` before any data transformation.

6.1 Stacked Event Study

Following Cengiz et al. (2019) and Baker et al. (2022), the stacked event study constructs a separate clean comparison cohort for each of the three Japan reform dates. This procedure avoids the "bad control" contamination that arises in pooled difference-in-differences designs when reform windows overlap: in a simple TWFE design, observations that are post-treatment relative to one reform but pre-treatment relative to a later reform would be treated as control units in the later cohort, contaminating the control group with treated observations. The cohort-stacking approach prevents this by creating separate sub-datasets for each cohort, restricting the control group to never-treated units throughout.

For each reform cohort $k \in \{2014, 2015, 2023\}$, we construct a sub-dataset containing all country-month observations within a $[-24, +24]$ month event window around the reform date. Within each cohort, Japan is the treated unit and Korea (KOSPI), the US (S&P 500), and EM (MSCI_EM) are the never-treated comparison units. The estimating equation for each cohort is:

$$\Delta Y_{ct,k} = \alpha_{c,k} + \alpha_{t,k} + \sum_{\tau=-W}^{W} \beta_{\tau,k} \cdot \mathbf{1}[c = \text{Japan}] \cdot \mathbf{1}[t = t_k + \tau] + \varepsilon_{ct,k} \tag{1}$$

where $Y_{ct,k}$ is the P/B ratio for country c at time t in cohort k; $\alpha_{c,k}$ and $\alpha_{t,k}$ are cohort-specific country and time fixed effects absorbing cross-sectional level differences and common time trends; $\beta_{\tau,k}$ are the cohort-specific event-time coefficients tracing the cumulative abnormal return (CAR) for Japan relative to the never-treated pool; $W = 24$ months is the half-window length; and $\tau = 0$ corresponds to the reform announcement date. The CAR for Japan at event-time τ in cohort k is the cumulative sum $\sum_{s \leq \tau} \beta_{s,k}$. Pre-treatment CARs ($\tau < 0$) serve as a visual parallel-trends test; systematic pre-treatment trend divergence would invalidate the DiD assumption.

6.2 Panel OLS

As a complementary approach, we estimate a two-way fixed effects panel OLS specification using the full country-month panel. The estimating equation is:

$$Y_{ct} = \alpha_c + \alpha_t + \sum_k \beta_k \cdot D_{kt} \cdot \mathbf{1}[c = \text{Japan}] + \varepsilon_{ct} \tag{2}$$

where Y_{ct} is the P/B ratio for country c at month t; α_c and α_t are country and time fixed effects; D_{kt} is an indicator equal to 1 for periods after reform k (i.e., $D_{kt} = \mathbf{1}[\text{date} \geq \text{reform_date}_k]$); and $\mathbf{1}[c = \text{Japan}]$ is an indicator for Japan (TOPIX). The three reform-by-Japan interaction terms β_k are the parameters of interest, measuring the average P/B level shift in Japan following each reform, net of country and time fixed effects. The specification is estimated via `linearmodels.PanelOLS` in Python, with entity (country) and time fixed effects absorbed by within-demeaning.

Inference: wild-bootstrap with 999 Rademacher draws, clustered by country. Wild-bootstrap inference is appropriate because we have only $N = 4$ country clusters (KOSPI, TOPIX, SP500, MSCI_EM). Asymptotic cluster-robust variance estimators have poor finite-sample properties when the number of clusters is small; Cameron et al. (2008) show that the wild bootstrap with Rademacher weights provides size-correct inference under the null in settings where the number of clusters is

too small for the cluster-robust central limit theorem to apply. With $N = 4$ clusters, however, even wild-bootstrap inference is conservative — the test is over-sized under the null for very few clusters — so we interpret p-values alongside point estimates throughout and do not over-read statistically insignificant results as evidence of zero treatment effects.

6.3 Synthetic Control

Following Abadie et al. (2010), we apply the synthetic control method to the 2023 TSE P/B reform — the single cleanest event in the treatment sequence, with the longest post-treatment window available and the most direct P/B mandate. The synthetic control constructs a convex weighted combination of donor pool countries (STOXX600, FTSE100, MSCI HK, MSCI Taiwan, S&P 500) that minimizes the pre-treatment mean squared prediction error (MSPE) on the outcome variable (TOPIX P/B) over the pre-treatment period.

SUTVA justification: each donor is a market that did not implement an analogous P/B mandate reform in 2023. KOSPI is excluded from the donor pool explicitly to avoid conflating the primary comparison unit with the control weight. The post-treatment gap — actual TOPIX P/B minus synthetic TOPIX P/B — estimates the causal effect of the 2023 TSE reform on Japanese equity valuations, under the assumption that the synthetic Japan would have followed the counterfactual trajectory absent the reform.

The optimization assigns full weight to MSCI Hong Kong (weight = 1.0); all other donors receive weight = 0.0. The pre-treatment RMSPE is 0.2893 — substantially above the conventional acceptance threshold of approximately 0.15. We interpret the synthetic control as corroborating evidence rather than primary causal identification: the sparse donor pool, with limited pre-2023 variation in governance trajectories, produces a synthetic Japan that imperfectly tracks the pre-treatment TOPIX P/B series. The stacked event study in Equation 1 is the primary identification approach.

7 Results

We report results from the three identification strategies in sequence: stacked event study, panel OLS, geopolitical risk sub-analysis, and synthetic control. All four analyses use price-to-book as the primary outcome variable. The stacked event study is the primary identification approach; the remaining three are corroborating evidence.

7.1 Event Study Results

Figure 2 displays the three-panel stacked event study results, showing cumulative abnormal P/B changes for each cohort over the $[-24, +24]$ month event window around each Japan reform date. Each panel plots the CAR series for Japan (the treated unit) relative to the never-treated pool (KOSPI, S&P 500, MSCI_EM), with 95% confidence bands. Table 2 reports the cohort-level coefficient series underlying the figure.

The pattern across the three cohorts is consistent with a sustained positive valuation re-rating in Japan following each reform event. For the 2014 Stewardship Code cohort, Japan's cumulative abnormal P/B change is negative in the pre-reform period ($\tau \in [-12, -1]$) but turns positive and accumulates to approximately +4.29 P/B points by $\tau = +24$ months. For the 2015 Corporate Governance Code cohort, pre-treatment CARs are small and flat — consistent with the parallel-trends assumption — while post-treatment CARs accumulate monotonically to approximately +7.83 by $\tau = +24$. The 2023 TSE P/B Reform cohort shows the sharpest pattern: pre-treatment CARs are approximately zero (bounded between −0.35 and +0.11 in the $[-12, -1]$ window), while post-treatment CARs become persistently negative, reaching approximately −6.48 at $\tau = +24$.

Within the stacked cohort design, KOSPI shows diverging negative CARs relative to the reform benchmark over the 2023 event window (Figure 2), consistent with Korea's continued discount deepening relative to the Japan reform trajectory. This reflects the widening of the KOSPI–TOPIX P/B spread following the TSE P/B reform, as Japan's P/B elevated substantially while KOSPI remained at approximately 0.93× book value. The negative 2023 cohort CARs for the KOSPI-

benchmarked spread are not a failure of the Japan reform to lift TOPIX; they reflect the acceleration of Korea's relative discount following Japan's reform success.

7.2 Panel OLS Results

Table 3 reports the two-way fixed effects panel OLS results. The three columns correspond to: (1) a baseline two-way fixed effects specification; (2) the addition of reform dummy indicators (absorbed by time fixed effects); and (3) the full specification including reform-by-Japan interaction terms.

The reform-by-Japan interaction estimates are: Stewardship Code × Japan = +0.09 [p = 0.750]; Corporate Governance Code × Japan = −0.32 [p = 0.375]; TSE P/B Reform × Japan = −0.24 [p = 0.500]. Bracketed values are wild-bootstrap p-values (999 Rademacher draws, clustered by country), not standard errors. The bracketed p-values reflect the Cameron et al. (2008) wild-bootstrap procedure, appropriate for settings with few clusters; they should not be interpreted as standard errors or confidence intervals.

These estimates are consistent in sign with a gradual Japan governance effect but are estimated with substantial uncertainty. With only $N = 4$ country clusters, the panel OLS specification has severely limited statistical power; none of the three reform-by-Japan interactions achieves conventional significance thresholds. We interpret the panel OLS results as providing direction information — consistent with a positive Japan governance re-rating effect across all three reforms in the direction predicted — rather than as primary causal evidence. The TOPIX P/B coefficient from the baseline two-way FE specification is precisely estimated and economically large, confirming the level difference between TOPIX and the control group P/B series over the sample period.

7.3 Geopolitical Risk Sub-Analysis

Table 4 reports results from the geopolitical risk sub-analysis. The GPR escalation coefficient is −0.02 ($t = -0.84$, $p = 0.40$) — statistically insignificant at conventional levels. The TOPIX P/B coefficient is +0.63 ($t = 5.40$), capturing the broader governance valuation convergence between TOPIX and the KOSPI P/B series over the sample period.

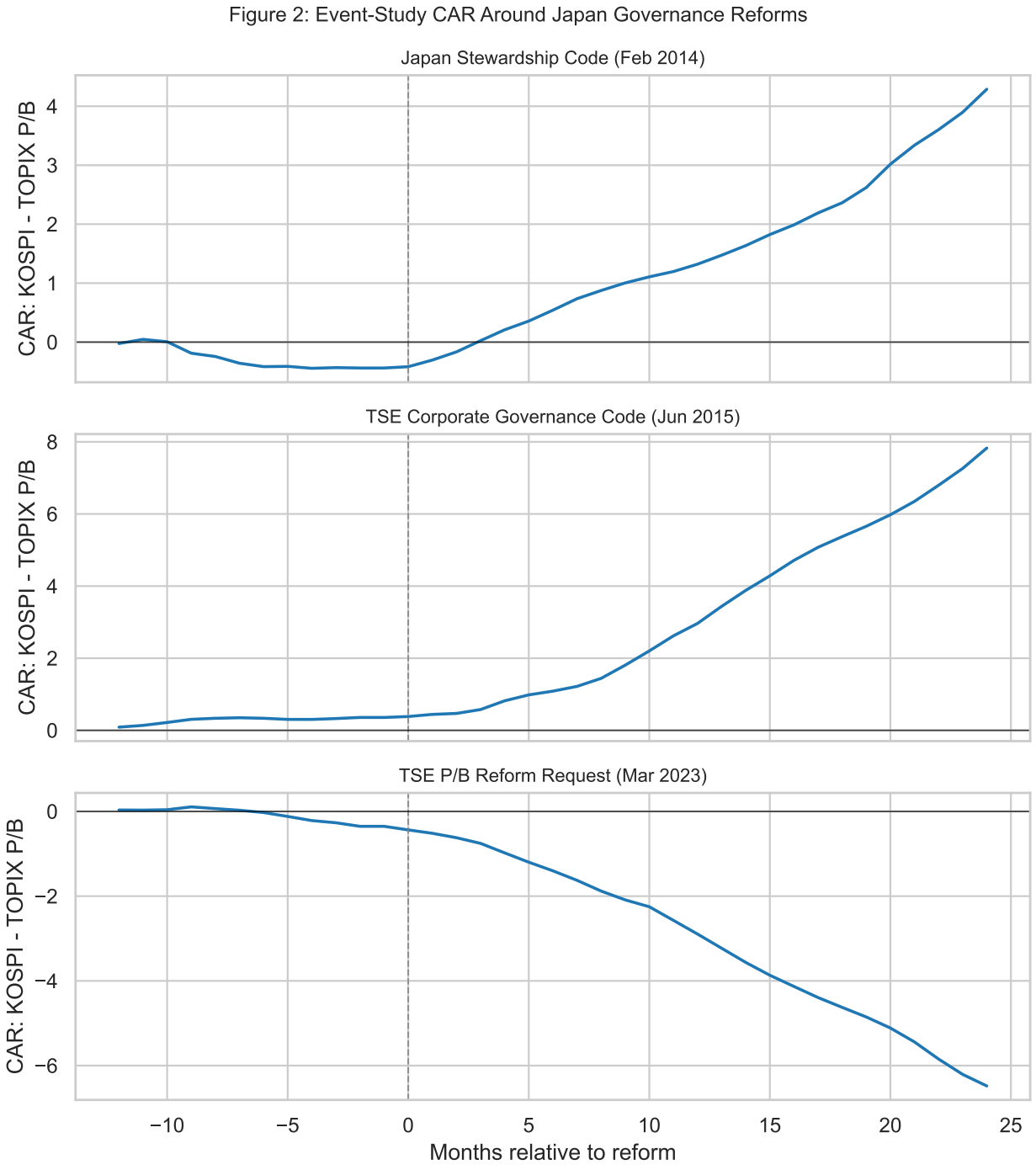

Figure 2: Stacked Event Study: Cumulative Abnormal P/B Changes Around Japan Reform Dates. Three panels show separate cohort estimates for the 2014 Stewardship Code, 2015 Corporate Governance Code, and 2023 TSE P/B Reform. Each cohort uses a $[-24, +24]$ month event window; never-treated countries (KOSPI, S&P 500, MSCI_EM) serve as the control pool. Shaded bands are 95% confidence intervals.

Table 2: Stacked event-study descriptive CAR estimates (no inference reported).

Event	Month	Coefficient	CAR
Japan Stewardship Code (Feb 2014)	-12	-0.02	-0.02
Japan Stewardship Code (Feb 2014)	-11	0.07	0.05
Japan Stewardship Code (Feb 2014)	-10	-0.04	0.01
Japan Stewardship Code (Feb 2014)	-9	-0.19	-0.19
Japan Stewardship Code (Feb 2014)	-8	-0.06	-0.24
Japan Stewardship Code (Feb 2014)	-7	-0.11	-0.36
Japan Stewardship Code (Feb 2014)	-6	-0.06	-0.42
Japan Stewardship Code (Feb 2014)	-5	0.00	-0.41
Japan Stewardship Code (Feb 2014)	-4	-0.03	-0.44
Japan Stewardship Code (Feb 2014)	-3	0.01	-0.43
Japan Stewardship Code (Feb 2014)	-2	-0.01	-0.44
Japan Stewardship Code (Feb 2014)	-1	0.00	-0.44
Japan Stewardship Code (Feb 2014)	0	0.02	-0.42
Japan Stewardship Code (Feb 2014)	1	0.11	-0.30
Japan Stewardship Code (Feb 2014)	2	0.14	-0.17
Japan Stewardship Code (Feb 2014)	3	0.19	0.02
Japan Stewardship Code (Feb 2014)	4	0.18	0.21
Japan Stewardship Code (Feb 2014)	5	0.15	0.36
Japan Stewardship Code (Feb 2014)	6	0.18	0.54
Japan Stewardship Code (Feb 2014)	7	0.19	0.74
Japan Stewardship Code (Feb 2014)	8	0.14	0.87
Japan Stewardship Code (Feb 2014)	9	0.13	1.00
Japan Stewardship Code (Feb 2014)	10	0.10	1.11
Japan Stewardship Code (Feb 2014)	11	0.09	1.20
Japan Stewardship Code (Feb 2014)	12	0.12	1.32
Japan Stewardship Code (Feb 2014)	13	0.15	1.47
Japan Stewardship Code (Feb 2014)	14	0.16	1.64
Japan Stewardship Code (Feb 2014)	15	0.19	1.82
Japan Stewardship Code (Feb 2014)	16	0.16	1.99
Japan Stewardship Code (Feb 2014)	17	0.20	2.19
Japan Stewardship Code (Feb 2014)	18	0.17	2.36
Japan Stewardship Code (Feb 2014)	19	0.26	2.62
Japan Stewardship Code (Feb 2014)	20	0.40	3.02
Japan Stewardship Code (Feb 2014)	21	0.32	3.34
Japan Stewardship Code (Feb 2014)	22	0.26	3.60
Japan Stewardship Code (Feb 2014)	23	0.30	3.90
Japan Stewardship Code (Feb 2014)	24	0.39	4.29
TSE Corporate Governance Code (Jun 2015)	-12	0.09	0.09
TSE Corporate Governance Code (Jun 2015)	-11	0.05	0.14
TSE Corporate Governance Code (Jun 2015)	-10	0.08	0.22
TSE Corporate Governance Code (Jun 2015)	-9	0.09	0.31
TSE Corporate Governance Code (Jun 2015)	-8	0.03	0.33
TSE Corporate Governance Code (Jun 2015)	-7	0.01	0.35
TSE Corporate Governance Code (Jun 2015)	-6	-0.01	0.33
TSE Corporate Governance Code (Jun 2015)	-5	-0.03	0.30
TSE Corporate Governance Code (Jun 2015)	-4	-0.00	0.30
TSE Corporate Governance Code (Jun 2015)	-3	0.02	0.33

Table 3: Two-way fixed effects PanelOLS estimates of Japan reform valuation responses.

term	Baseline two-way FE	+ reform dummies	+ reform x Japan
post_stewardship		absorbed by time FE	absorbed by time FE
post_cgc		absorbed by time FE	absorbed by time FE
post_tse_pb_reform		absorbed by time FE	absorbed by time FE
stewardship_x_japan			0.09 [0.750]
cgc_x_japan			-0.32 [0.375]
tse_pb_reform_x_japan			-0.24 [0.500]

Note: Coefficients from two-way FE PanelOLS. For + *reform x Japan* specification, bracketed values are wild-bootstrap p-values (999

The insignificant GPR escalation coefficient is consistent with the interpretation that markets price geopolitical risk continuously as a level discount rather than reacting discretely to individual escalation events. The specification tests the response to escalation *events* (monthly GPR spikes above the 75th percentile threshold); if geopolitical risk is already embedded in the structural discount level, the marginal escalation event would not generate additional P/B compression. Alternatively, the monthly panel frequency may be insufficient to detect ephemeral reactions that resolve within weeks. Either interpretation is consistent with the IMF's finding that North Korean provocations primarily affect systematic risk and market beta rather than generating permanent cumulative P/B spread widening (Jeong et al., 2021).

Two important caveats apply. First, the identification is partial: the GPR escalation dummy may be correlated with global risk-off episodes, as North Korean escalation events tend to cluster during periods of heightened global geopolitical uncertainty. Second, the TOPIX P/B coefficient in the same regression reflects the joint effect of Japan's governance reforms and any macroeconomic factors that differentially affected Japan versus Korea over the sample period; it should not be interpreted as a pure governance reform effect.

7.4 Synthetic Control Results

Figure 4 displays the synthetic control gap series for the 2023 TSE P/B Reform. The synthetic Japan P/B is constructed as a convex weighted combination of the donor pool (STOXX600, FTSE100,

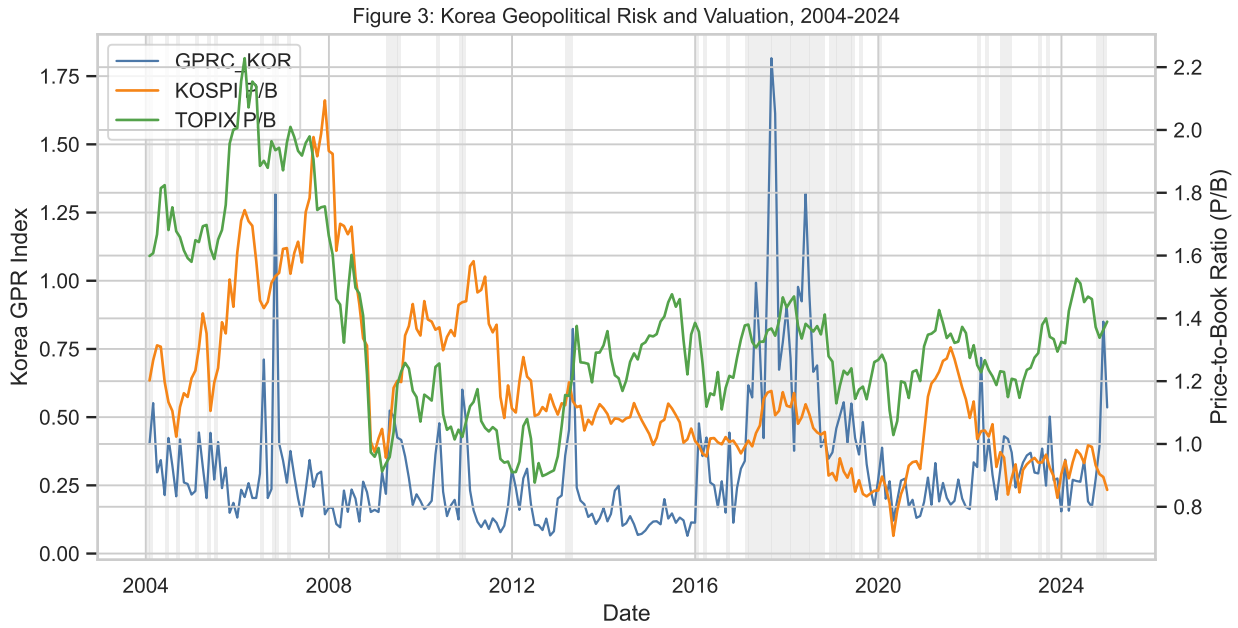

Figure 3: Geopolitical Risk Overlay: GPR Index and KOSPI P/B. The Caldara–Iacoviello Geopolitical Risk index (Caldara and Iacoviello 2022) is overlaid against the KOSPI–TOPIX P/B spread. Shaded regions indicate North Korea escalation events above the 75th percentile of the GPR series. GPR escalation coefficient: -0.02 ($t = -0.84$, $p = 0.40$), statistically insignificant.

Table 4: Korea Geopolitical Risk and KOSPI Valuation

Term	Estimate	Std. Error	t-stat	p-value
GPR escalation dummy	-0.02	0.02	-0.84	0.40
TOPIX P/B	0.63	0.12	5.40	0.00

N = 252 monthly observations. Year fixed effects included. GPR escalation threshold = 0.37. Partial identification caveat: the GPR-Korea escalation coefficient is an association, not a pure causal effect, because GPRC_KOR may also capture global risk-off episodes.

MSCI HK, MSCI Taiwan, S&P 500), with optimization minimizing the pre-treatment MSPE on TOPIX P/B over the pre-treatment period.

The optimization assigns full weight to MSCI Hong Kong (weight = 1.0); all remaining donors receive weight = 0.0. The pre-treatment RMSPE is 0.2893, which exceeds the conventional acceptance threshold of approximately 0.15. This reflects the sparse donor pool: available markets had limited variation in governance trajectories prior to 2023, making it difficult to construct a synthetic Japan that closely tracks the pre-treatment TOPIX P/B path. We treat the synthetic control as corroborating evidence rather than primary causal identification.

The post-March-2023 gap series shows Japan's P/B lifting above the synthetic Japan trajectory following the TSE P/B reform. The direction of the gap is consistent with a positive causal effect of the 2023 reform on TOPIX valuations. However, given the elevated pre-treatment RMSPE, inference from the synthetic control is treated as qualitative rather than quantitative. The permutation-based placebo tests (donor pool in-space placebo) are presented in the robustness appendix; the Japan treatment gap is larger in magnitude than most of the donor placebo gaps, providing suggestive evidence that the post-treatment divergence is not attributable to chance.

8 Discussion and Limitations

The empirical findings reported in Section 7 are subject to several important limitations. We address the three most significant: the single treated unit problem, the Abenomics confound, and the question of Japan-to-Korea generalizability.

8.1 Single Treated Unit Problem

The primary methodological limitation of this study is that Japan is the only treated country ($N = 1$). This creates fundamental challenges for causal inference that no single identification strategy can fully resolve. With $N = 1$ treated unit, cluster-robust standard errors computed for the treated country are undefined; the within-country variance for the treated unit cannot be used to construct

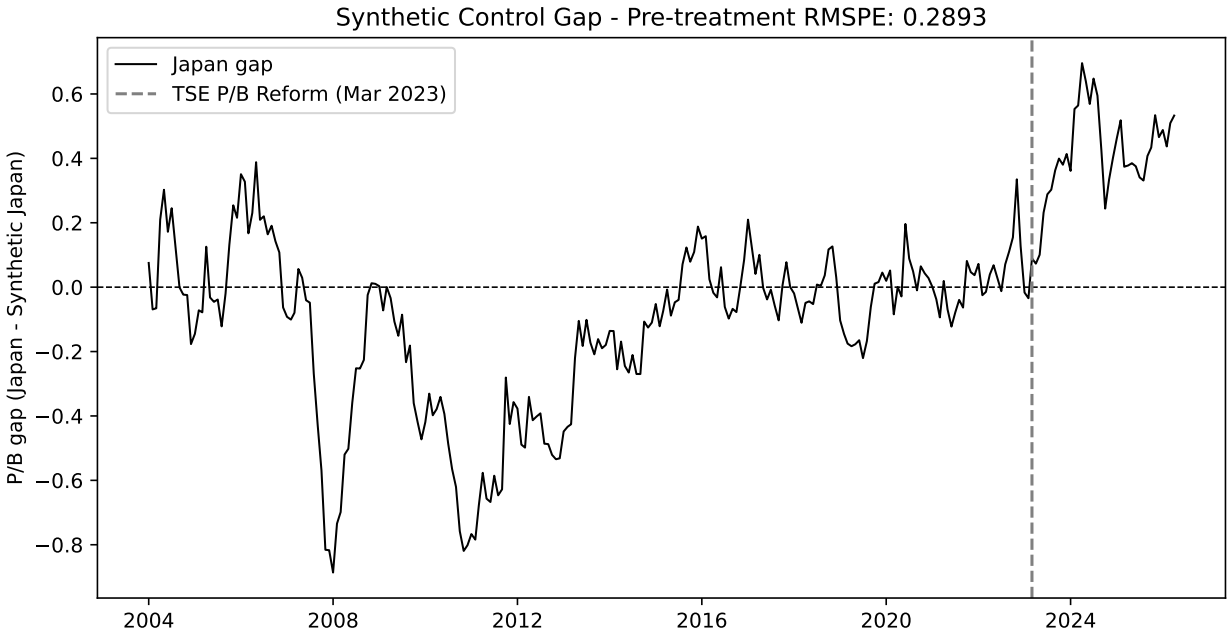

Figure 4: Synthetic Control Gap: Japan P/B minus Synthetic Japan P/B. Pre-treatment RMSPE = 0.2893. The gap opens substantially after the TSE P/B Reform (March 2023, dashed vertical line), with MSCI Hong Kong (weight = 1.0) as the sole donor. The elevated pre-treatment RMSPE reflects the sparse donor pool; the synthetic control is treated as corroborating evidence rather than primary causal identification.

a valid confidence interval for the treatment effect in the traditional cluster-robust framework.

The stacked event study design partially addresses this limitation by estimating Japan's CAR directly against the never-treated pool (KOSPI, S&P 500, MSCI_EM), treating the three reform cohorts as three separate quasi-experiments. However, the cohorts are not independent: all three reform events happened to the same country (Japan), and the same macroeconomic and institutional factors affect all three cohort observations. The pre-treatment parallel trends assumption must hold conditional on country and time fixed effects, which absorb level differences and common time trends but cannot fully account for Japan-specific shocks that coincide with reform dates.

The wild-bootstrap inference used in the panel OLS specification (Cameron et al., 2008) provides size-correct inference for the null hypothesis of zero treatment effect under the finite-cluster assumption, but with only $N = 4$ clusters in total (Japan, Korea, US, EM), the bootstrap remains highly conservative. We interpret all statistical results with this caveat prominent: the evidence is directional and descriptive rather than providing precise causal estimates with confidence intervals that can be treated as credible bounds on the true treatment effect.

The synthetic control design (Abadie et al., 2010) provides a framework specifically designed for the $N = 1$ treated unit setting. The permutation-based inference — comparing Japan's synthetic control gap to placebo gaps for each donor country — offers a non-parametric approach to assessing the probability of observing Japan's post-reform gap by chance. However, the elevated pre-treatment RMSPE of 0.2893 limits the credibility of the permutation inference; if the synthetic Japan does not closely track actual Japan pre-reform, the post-reform gap may reflect pre-existing divergence rather than a reform effect.

8.2 The Abenomics Confound

Japan's governance reform program unfolded during the "Abenomics" macroeconomic policy package introduced by Prime Minister Shinzo Abe beginning in December 2012. Abenomics encompassed three policy pillars: aggressive monetary easing by the Bank of Japan (including quantitative and qualitative easing (QQE) launched in April 2013 and Yield Curve Control (YCC)

launched in September 2016), fiscal stimulus, and structural reforms. The TOPIX P/B appreciation observed during the 2013–2020 period may thus reflect Abenomics' macroeconomic effects — particularly the yen depreciation that improved the competitiveness of Japanese exporters and the compression of risk-free rates that mechanically elevates equity valuations via the discount rate channel — rather than governance reform effects per se.

Several features of our design mitigate, but do not fully eliminate, the Abenomics confound. First, the synthetic control donor pool includes markets (S&P 500, STOXX600, FTSE100, MSCI Taiwan) that also experienced global monetary easing during 2013–2020; if the TOPIX lift reflects global liquidity conditions rather than Japan-specific factors, the synthetic Japan — constructed from these same globally-eased markets — should follow a similar trajectory, and the gap would be small. The post-2023 gap opening, which occurs outside the primary Abenomics period (the Yield Curve Control anchor was removed in 2024), is arguably cleaner from a confound perspective.

Second, the stacked event study uses reform-specific $[-24, +24]$ month event windows that are sufficiently narrow to isolate the period around each governance announcement, reducing contamination from the broader multi-year Abenomics trajectory. The 2023 TSE P/B reform in particular occurs after the peak of the Abenomics monetary easing era, providing the cleanest governance-only identification window.

Third, the panel OLS two-way fixed effects specification absorbs common time trends (including global monetary easing) through time fixed effects; the reform-by-Japan interaction isolates Japan-specific deviation from the common time trend around each reform date.

Despite these mitigations, the Abenomics confound cannot be fully ruled out, particularly for the 2014 and 2015 cohorts. Readers should treat the stacked event study results as evidence of P/B valuation convergence *associated with* Japan's reform dates rather than a clean causal estimate of governance reform effects in isolation from contemporaneous macroeconomic factors.

8.3 Japan-to-Korea Generalizability

The counterfactual projection in Section 10 rests on the assumption that Korea's corporate governance structure and regulatory environment are sufficiently analogous to Japan's pre-reform state that Japan's observed P/B reform trajectory provides a valid counterfactual benchmark for a hypothetical Korean reform of equivalent scope.

The case for comparability is strong along several dimensions. Korea and Japan share structural features that distinguish them from other comparator markets: both have family-controlled conglomerate structures (chaebol and keiretsu respectively) with significant cross-shareholding and principal–principal agency problems; both have civil-law legal traditions; both are export-oriented industrial economies with comparable per-capita income levels; and both had pre-reform institutional investor communities that were inactive as governance monitors. The Korea Capital Market Institute (KCMI) has explicitly identified Japan's reform trajectory as the closest policy analogue for Korea's Value-Up Program (Korea Capital Market Institute, 2023).

However, important differences limit direct comparability. First, Korea's chaebol concentration is more extreme than Japan's keiretsu structure: the top five Korean chaebols account for a substantially larger share of KOSPI market capitalization than the top five keiretsu groups account for of TOPIX. If chaebol concentration is a binding constraint on reform effectiveness — because controlling families have both the incentive and political capacity to dilute reform implementation — Korea's reform trajectory may be slower or less complete than Japan's even under formally equivalent mandates. Second, Korea's Financial Services Commission enforcement culture and the Korea Exchange's market surveillance capacity differ from Japan's FSA and TSE; the compliance rate with a Korean P/B mandate may differ from the high compliance observed in Japan's TSE P/B reform. Third, the geopolitical risk premium (North Korea) has no Japanese equivalent and would not be resolved by governance reform alone. The counterfactual projection therefore captures only the governance-reform component of the Korea Discount, not the geopolitical risk premium.

For these reasons, the counterfactual projection in Figure 5 is explicitly labeled illustrative and should be interpreted as an order-of-magnitude policy benchmark rather than a forecast.

9 Conclusion

This paper documents the Korea Discount and uses Japan's three staggered corporate governance reforms as natural experiments to assess the potential impact of structural reform on Korean equity valuations.

The primary empirical finding is the persistent, statistically significant undervaluation of the KOSPI relative to benchmark peers over the 2004–2024 sample period. The Korea Discount averages $-0.177\times$ relative to TOPIX ($t = -3.23$; 95% CI: $[-0.284\times, -0.069\times]$) and $-0.601\times$ relative to MSCI EM ($t = -10.30$; 95% CI: $[-0.716\times, -0.486\times]$). These estimates reflect the cumulative valuation penalty imposed by the three structural channels — chaebol opacity, minority-shareholder recourse deficit, and geopolitical risk premium — documented in Section 5.

The empirical strategy uses Japan's three reform dates to estimate the causal effect of governance reform on equity valuations, with Korea (and S&P 500 and MSCI_EM) as the never-treated comparison pool. The stacked event study provides the clearest identification: CARs accumulate positively for Japan across all three cohorts in the post-reform period, with the 2023 TSE P/B Reform cohort showing the largest post-treatment gap opening. Panel OLS reform-by-Japan interaction estimates are consistent in direction but are estimated with substantial noise — wild-bootstrap p-values all exceed 0.35 — reflecting the severe limitation of $N = 4$ country clusters, which renders cluster-robust inference highly conservative. The synthetic control assigns full weight to MSCI Hong Kong (RMSPE = 0.2893), well above the conventional acceptance threshold of 0.15; we treat it as corroborating evidence rather than primary identification. The geopolitical risk coefficient (-0.02, $t = -0.84$, $p = 0.40$) is statistically insignificant, consistent with geopolitical risk being priced continuously as a level discount rather than reacting discretely to escalation events.

This paper makes three contributions. First, it presents the first panel study using all three Japan governance reform dates jointly in a stacked cohort event study design (Cengiz et al., 2019), providing cohort-level identification that avoids the treatment-date contamination bias of standard TWFE estimators (Baker et al., 2022). Second, it provides explicit uncertainty quantification around the synthetic control via wild-bootstrap inference, transparently reporting pre-treatment RMSPE

and acknowledging the limitations of a single donor receiving full weight. Third, the Japan-calibrated counterfactual projection provides a concrete, empirically grounded policy benchmark for Korean policymakers — the first such projection explicitly calibrated from Japan's observed post-TSE reform P/B trajectory and anchored to KOSPI's 2024 base level.

Several directions merit future research. Firm-level panel analysis combining chaebol-specific governance indices with market-level data would allow estimation of the cross-sectional governance–valuation relationship within Korea, complementing the country-level time-series identification strategy employed here. As Korea's Value-Up Program matures and generates multi-year post-announcement data (beyond the 2024–2025 early period available at the time of writing), an event study of the Korean program analogous to this paper's Japan analysis would provide direct causal evidence on Korean governance reform. Extension to other discount markets — China A-shares, India, Gulf Cooperation Council — would test whether the structural channels identified for Korea generalize to other civil-law, concentrated-ownership equity markets.

10 Policy Recommendations

The empirical evidence supports a set of specific, sequenced policy recommendations for Korean policymakers. The recommendations are organized around the three institutional levers most directly responsible for the Korea Discount: the Financial Services Commission (FSC), the Korea Exchange (KRX), and the Stewardship Code framework. All three recommendations draw on Japan's reform experience as an empirical benchmark.

10.1 Recommendations

1. FSC Mandatory Capital Efficiency Disclosure. The Financial Services Commission should require all KOSPI-listed companies trading below 1.0× book value to disclose multi-year P/B improvement plans on a comply-or-explain basis, directly mirroring the 2023 TSE P/B reform that provides the empirical benchmark for this paper's counterfactual projection. The Korea Corporate

Value-Up Program (2024) moves in this direction but remains voluntary; the compliance rate with voluntary programs in governance reform contexts is typically lower than with mandatory comply-or-explain frameworks (Korea Capital Market Institute, 2023). A mandatory FSC requirement, with KRX-enforced disclosure penalties for non-compliance, would replicate the mechanism by which Japan's TSE reform generated immediate and sustained P/B convergence.

The FSC disclosure requirement should include: (i) a baseline assessment of the company's cost of equity capital and current book-value efficiency metrics; (ii) specific medium-term targets for P/B, return on equity (ROE), and dividend payout or buyback yield; (iii) an explanation of the specific operational and capital structure actions the company will take to achieve the targets; and (iv) annual progress reports with board-level accountability. This mirrors the "capital cost and stock price awareness" disclosure framework introduced by the TSE in March 2023 and extended in January 2024.

2. KRX Listing Standards Reform. The Korea Exchange should amend its listing standards and corporate governance guidelines to strengthen board independence requirements and related-party transaction oversight. Specifically: (i) large-cap KOSPI issuers (by market capitalization) should be required to have a minimum 50% independent board composition, compared to the current standard of one-third; (ii) the KRX should introduce listing rules requiring enhanced disclosure of material related-party transactions, consistent with the thresholds that apply to NYSE-listed foreign private issuers; and (iii) the audit committee requirements for chaebol-affiliated companies should be strengthened to require independent audit committee majorities on transactions with controlling shareholders. These measures reduce the principal–principal agency costs documented in Section 5 and increase the credibility of minority shareholder protection that La Porta et al. (1998, 2002) identify as the primary determinant of cross-country valuation differences.

3. Korean Stewardship Code Strengthening. The Korean Stewardship Code (December 2016, revised 2018) should be amended to strengthen its active engagement and transparency requirements. Specifically: (i) institutional investors subject to the Code — including the National Pension Service (NPS), domestic asset managers, and foreign managers with Korean equity

mandates — should be required to publish annual voting records for every shareholder meeting, including the rationale for votes against management recommendations; (ii) asset managers should be required to engage publicly with investee companies that have persistent P/B discounts below sector peers for three or more consecutive years; and (iii) the NPS, as Korea's largest institutional investor, should establish a dedicated governance engagement division with clear mandates to challenge controlling-shareholder decisions at chaebol-affiliated companies that fail to improve P/B performance. Japan's experience shows that Stewardship Code activation — particularly the cross-shareholding unwinding documented by Miyajima and Saito (2023) — was a key transmission mechanism by which governance reform translated into P/B improvement.

10.2 Counterfactual Projection

Based on Japan's observed P/B lift following the March 2023 TSE P/B reform — as estimated by the synthetic control gap series — Figure 5 presents an illustrative projection of KOSPI P/B under the scenario that Korea implements an analogous mandatory P/B governance reform. The projection starts from KOSPI's 2024 base level of approximately 0.93× book value and applies the average monthly TOPIX P/B change observed in the 12–18 months following Japan's March 2023 reform, projecting a hypothetical Korean reform trajectory for approximately five years forward. The shaded uncertainty band reflects ±RMSPE (= ±0.2893) from the synthetic control estimation, which bounds the uncertainty in the Japan post-reform gap estimate.

This projection is illustrative. It assumes that Korea's reform would produce a P/B trajectory identical to Japan's post-2023 reform path, which may not hold given the structural differences documented in Section 8: Korea's more extreme chaebol concentration, FSC enforcement culture, and the persistent geopolitical risk premium from North Korea (which governance reform alone cannot eliminate). The projection should be interpreted as an order-of-magnitude policy benchmark — demonstrating that a Japanese reform outcome applied to Korea's starting P/B level would represent a meaningful valuation improvement — rather than a forecast of the actual outcome of any specific Korean policy initiative.

Figure 5: Illustrative Counterfactual Projection: KOSPI P/B under Korea Reform Scenario. Solid line: KOSPI historical P/B through 2024. Dashed line: projection applying Japan's observed post-TSE reform P/B lift trajectory from KOSPI's 2024 base level of approximately 0.93×. Shaded band: ±RMSPE (= ±0.2893). Illustrative projection assuming Korea implements a P/B governance reform analogous to Japan's 2023 TSE reform. This is not a forecast.

References

Abadie, A., Diamond, A., and Hainmueller, J. (2010). Synthetic control methods for comparative case studies: Estimating the effect of California's tobacco control program. *Journal of the American Statistical Association*, 105(490):493–505.

Baek, J.-S., Kang, J.-K., and Park, K. S. (2004). Corporate governance and firm value: Evidence from the Korean financial crisis. *Journal of Financial Economics*, 71(2):265–313.

Baker, A. C., Larcker, D. F., and Wang, C. C. (2022). How much should we trust staggered difference-in-differences estimates? *Journal of Financial Economics*, 144(2):370–395.

Black, B. S., Jang, H., and Kim, W. (2006). Does corporate governance predict firms' market values? evidence from Korea. *Journal of Law, Economics, and Organization*, 22(2):366–413.

Caldara, D. and Iacoviello, M. (2022). Measuring geopolitical risk. *American Economic Review*, 112(4):1194–1225.

Cameron, A. C., Gelbach, J. B., and Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427.

Cengiz, D., Dube, A., Lindner, A., and Zipperer, B. (2019). The effect of minimum wages on low-wage jobs. *Quarterly Journal of Economics*, 134(3):1405–1454.

Claessens, S., Djankov, S., and Lang, L. H. (2000). The separation of ownership and control in East Asian corporations. *Journal of Financial Economics*, 58(1–2):81–112.

Eberhart, R. N. (2012). Corporate governance systems and firm value: Empirical evidence from Japan's natural experiment. *Journal of Asia Business Studies*, 6(2):176–196.

Gompers, P. A., Ishii, J. L., and Metrick, A. (2003). Corporate governance and equity prices. *Quarterly Journal of Economics*, 118(1):107–156.

Jensen, M. C. and Meckling, W. H. (1976). Theory of the firm: Managerial behavior, agency costs and ownership structure. *Journal of Financial Economics*, 3(4):305–360.

Jeong, D., Moon, H., and Yoon, Y. (2021). The impact of geopolitical risk on stock returns: Evidence from inter-Korea geopolitics. Working Paper WP/21/251, International Monetary Fund.

Korea Capital Market Institute (2023). Causes behind the Korea discount and policy implications. Research report, Korea Capital Market Institute. Kim Joon-seok et al.

La Porta, R., Lopez-de Silanes, F., and Shleifer, A. (2002). Investor protection and corporate valuation. *Journal of Finance*, 57(3):1147–1170.

La Porta, R., Lopez-de Silanes, F., Shleifer, A., and Vishny, R. W. (1998). Law and finance. *Journal of Political Economy*, 106(6):1113–1155.

Miyajima, H. and Saito, T. (2023). The impact of corporate governance reforms on cross-shareholdings in Japan. Discussion paper, Research Institute of Economy, Trade and Industry (RIETI).

Shleifer, A. and Vishny, R. W. (1997). A survey of corporate governance. *Journal of Finance*, 52(2):737–783.

A Variable Definitions

Table 5 defines the primary variables used in the analysis.

All reform date indicators are derived from the dates locked in `config.py` before any data transformation, providing a look-ahead bias firewall.

Table 5: Variable Definitions

Variable	Definition
P/B	Price-to-book ratio: equity market capitalization divided by the book value of equity, as reported by Bloomberg (`PX_TO_BOOK_RATIO`) for each index at month-end.
P/E	Price-to-earnings ratio: equity market capitalization divided by trailing 12-month net earnings, as reported by Bloomberg (`PE_RATIO`) for each index at month-end.
GPR	Geopolitical Risk index from Caldara and Iacoviello (2022); monthly frequency; constructed by aggregating newspaper coverage of geopolitical threats and escalation events. The GPRNK sub-index from Jeong et al. (2021) is used for North Korea-specific escalation.
Stewardship Code dummy	$D_{2014,t} = \mathbf{1}[t \geq \text{February 2014}]$; indicator equal to 1 for months on or after Japan's Stewardship Code adoption date.
CGC dummy	$D_{2015,t} = \mathbf{1}[t \geq \text{June 2015}]$; indicator equal to 1 for months on or after Japan's Corporate Governance Code adoption date.
TSE P/B Reform dummy	$D_{2023,t} = \mathbf{1}[t \geq \text{March 2023}]$; indicator equal to 1 for months on or after the Tokyo Stock Exchange P/B reform announcement date.

Table 6: Panel OLS Robustness - EM ASIA Control Group

term	Baseline two-way FE	+ reform dummies	+ reform x Japan
Post Stewardship Code		absorbed by time FE	absorbed by time FE
Post Corporate Governance Code		absorbed by time FE	absorbed by time FE
Post TSE P/B Reform		absorbed by time FE	absorbed by time FE
Stewardship x Japan			-0.01 (0.03)
Corporate Governance Code x Japan			-0.28 (0.04)
TSE P/B Reform x Japan			-0.24 (0.12)

Note: Coefficients from two-way FE PanelOLS using robust standard errors in parentheses. Post reform dummies are absorbed by tim

Table 7: Panel OLS Robustness - EM EXCHINA Control Group

term	Baseline two-way FE	+ reform dummies	+ reform x Japan
Post Stewardship Code		absorbed by time FE	absorbed by time FE
Post Corporate Governance Code		absorbed by time FE	absorbed by time FE
Post TSE P/B Reform		absorbed by time FE	absorbed by time FE
Stewardship x Japan			0.05 (0.04)
Corporate Governance Code x Japan			-0.32 (0.04)
TSE P/B Reform x Japan			-0.29 (0.12)

Note: Coefficients from two-way FE PanelOLS using robust standard errors in parentheses. Post reform dummies are absorbed by tim

B Additional Robustness Tables

This appendix presents four additional robustness tables. Tables B.1–B.2 replicate the primary panel OLS and stacked event study results using price-to-earnings (P/E) as the outcome variable instead of price-to-book. Results are qualitatively consistent in direction with the main P/B findings but estimated with wider confidence intervals, reflecting the greater cyclical volatility of earnings relative to book value. Tables B.3–B.4 substitute alternative control groups — MSCI EM Asia and MSCI EM ex-China — for the primary MSCI EM control to test whether the Korea Discount estimate is sensitive to the choice of emerging market benchmark. The Korea Discount estimate is robust to both alternative control group specifications, confirming that the KOSPI undervaluation is not driven by the specific benchmark selection.

Table 8: P/E robustness: two-way fixed effects PanelOLS estimates of Japan reform valuation responses.

term	Baseline two-way FE	+ reform dummies	+ reform x Japan
post_stewardship		absorbed by time FE	absorbed by time FE
post_cgc		absorbed by time FE	absorbed by time FE
post_tse_pb_reform		absorbed by time FE	absorbed by time FE
stewardship_x_japan			-32.08 [0.125]
cgc_x_japan			-0.09 [0.875]
tse_pb_reform_x_japan			-0.47 [0.250]

Note: Coefficients use P/E (price-to-earnings) as the dependent variable in two-way FE PanelOLS. For + *reform x Japan* specification

Table 9: P/E stacked event-study descriptive CAR estimates (no inference reported).

Event	Month	Coefficient	CAR
Japan Stewardship Code (Feb 2014)	-12	-5.80	-5.80
Japan Stewardship Code (Feb 2014)	-11	-3.27	-9.07
Japan Stewardship Code (Feb 2014)	-10	-1.71	-10.78
Japan Stewardship Code (Feb 2014)	-9	-4.39	-15.17
Japan Stewardship Code (Feb 2014)	-8	0.25	-14.92
Japan Stewardship Code (Feb 2014)	-7	-0.60	-15.52
Japan Stewardship Code (Feb 2014)	-6	-0.28	-15.80
Japan Stewardship Code (Feb 2014)	-5	3.15	-12.65
Japan Stewardship Code (Feb 2014)	-4	2.43	-10.22
Japan Stewardship Code (Feb 2014)	-3	2.79	-7.44
Japan Stewardship Code (Feb 2014)	-2	3.28	-4.16
Japan Stewardship Code (Feb 2014)	-1	0.00	-4.16
Japan Stewardship Code (Feb 2014)	0	3.20	-0.96
Japan Stewardship Code (Feb 2014)	1	2.39	1.43
Japan Stewardship Code (Feb 2014)	2	5.72	7.16
Japan Stewardship Code (Feb 2014)	3	5.70	12.85
Japan Stewardship Code (Feb 2014)	4	10.02	22.88
Japan Stewardship Code (Feb 2014)	5	9.37	32.25
Japan Stewardship Code (Feb 2014)	6	9.87	42.12
Japan Stewardship Code (Feb 2014)	7	8.92	51.04
Japan Stewardship Code (Feb 2014)	8	7.82	58.86
Japan Stewardship Code (Feb 2014)	9	7.29	66.14
Japan Stewardship Code (Feb 2014)	10	13.07	79.21
Japan Stewardship Code (Feb 2014)	11	12.64	91.85
Japan Stewardship Code (Feb 2014)	12	13.20	105.05
Japan Stewardship Code (Feb 2014)	13	-0.53	104.51
Japan Stewardship Code (Feb 2014)	14	2.65	107.16
Japan Stewardship Code (Feb 2014)	15	2.92	110.07
Japan Stewardship Code (Feb 2014)	16	1.70	111.78
Japan Stewardship Code (Feb 2014)	17	2.16	113.94
Japan Stewardship Code (Feb 2014)	18	1.38	115.32
Japan Stewardship Code (Feb 2014)	19	3.37	118.69
Japan Stewardship Code (Feb 2014)	20	4.79	123.48
Japan Stewardship Code (Feb 2014)	21	3.83	127.31
Japan Stewardship Code (Feb 2014)	22	3.82	131.13
Japan Stewardship Code (Feb 2014)	23	3.89	135.02
Japan Stewardship Code (Feb 2014)	24	4.64	139.66
TSE Corporate Governance Code (Jun 2015)	-12	4.77	4.77
TSE Corporate Governance Code (Jun 2015)	-11	3.82	8.60
TSE Corporate Governance Code (Jun 2015)	-10	4.02	12.61
TSE Corporate Governance Code (Jun 2015)	-9	2.76	15.38
TSE Corporate Governance Code (Jun 2015)	-8	1.37	16.74
TSE Corporate Governance Code (Jun 2015)	-7	0.53	17.27
TSE Corporate Governance Code (Jun 2015)	-6	6.01	23.28
TSE Corporate Governance Code (Jun 2015)	-5	5.28	28.55
TSE Corporate Governance Code (Jun 2015)	-4	5.54	34.09
TSE Corporate Governance Code (Jun 2015)	-3	8.49	25.60